Constructing Conformal Predictive Distributions via Kernel-Localized Residuals

From Local Empirical Distributions to Rank-Based Sampling

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Project Objective

Project Objective: From Point Estimates to Distributions

Conditional Mean

$Y \approx \mathbb{E}[Y \mid X]$

- Regression
- ML Models

→

Predictive Interval

$Y \in \Gamma_{1-\alpha}(x)$

- Conformal Prediction

→

Predictive Distribution

$Y \sim \hat{F}(\cdot|x)$

(Our Goal)

Moving from a single point estimate to a full predictive distribution provides the richest uncertainty quantification, but is more challenging.

Why Conditional Distributions Matter

Predicting a full conditional distribution enables:

- **Decision making**: expected-utility computation for risk-aware choices.
- **Electrical load forecasting**: uncertainty-aware operations planning.
- **Digital soil mapping**: uncertainty maps for scientific decisions.
- **Volatility prediction**: risk quantification in finance.

Message: Accurate conditional distributions are necessary for reliable decisions.

Methods for Estimating Conditional Distributions

Tree-based methods

Classification and Regression Tree (CART) (Reiter, 2005), Quantile Regression Forests (Meinshausen, 2006)

Deep learning based approachs

Generative Adversarial Networks (Xu et al., 2019), Diffusion Models (Kotelnikov et al., 2023), Mixture Density Networks (Bishop, 1994)

A Common Limitation

These methods can approximate conditional distributions well, but they do not provide **distribution-free, finite-sample assurances** for new data.

Conformal Prediction: Trustworthy Uncertainty

Paradigm: With **exchangeability** only, conformal prediction yields prediction sets $\Gamma_{1-\alpha}(X_*)$ with finite-sample coverage (Vovk et al., 2005).

$$\mathbb{P}\{Y_* \in \Gamma_{1-\alpha}(X_*)\} \geq 1 - \alpha.$$

General (Split) Conformal Prediction Set Construction

1. **Split data** into training set $\mathcal{T}$ and calibration set $\mathcal{C}$.
2. **Learn score function** $s(x, y)$ using only $\mathcal{T}$. (This $s(x, y)$ measures the "non-conformity" or "strangeness" of y given x).
3. **Compute calibration scores**: $S_i = s(X_i, Y_i)$ for all $i \in \mathcal{C}$.
4. **Find quantile**: $\hat{q}$ is the $\lceil (1-\alpha)(|\mathcal{C}|+1) \rceil / (|\mathcal{C}|+1)$ quantile of the scores $\{S_i\}$.
5. **Construct set**: $\Gamma_{1-\alpha}(x) = \{y \in \mathbb{R} : s(x, y) \leq \hat{q}\}$.

Split-Conformal Predictive Systems (SCPS)

Recent work on conformal prediction has been extended from intervals to **full predictive distributions** with finite-sample marginal coverage.

Split-Conformal Predictive Systems (SCPS) provide one way to generate these predictive distributions (Vovk et al., 2020).

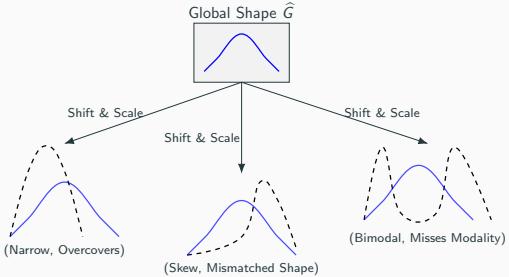
SCPS Mechanism

1. **Split data** into training set $\mathcal{T}$ and calibration set $\mathcal{C}$.
2. **Estimate** the conditional mean $\hat{\mu}$ and the scale $\hat{\sigma}$ on $\mathcal{T}$.
3. **Compute standardized residuals on** $\mathcal{C}$: $Z_i = (Y_i - \hat{\mu}(X_i))/\hat{\sigma}(X_i)$ for $i \in \mathcal{C}$.
4. **Build a single empirical CDF,** $\widehat{G}$**,** from all $\{Z_i\}$.
5. At test point x_*, **the predictive CDF is** $\widehat{G}\left(\frac{y-\hat{\mu}(x_*)}{\hat{\sigma}(x_*)}\right)$.

Limitation: One Global Shape Cannot Serve All x

A key limitation of SCPS is that reusing a *single* global residual CDF $\widehat{G}$ for all test points x:

- **Ignores** how the true error distribution's *shape* (skew, modality) changes with x.
- **Fails** to capture this local structure
- **Becomes** an inaccurate approximation when the residual shape is not x-invariant.



SCPS applies the same global shape (blue) everywhere, failing to match true local distributions (dashed black).

Our Goal

Goal: To construct a predictive distribution that reflects the **local characteristics** of the true conditional distribution, while **offering distribution-free coverage guarantees**.

Approach: Developed within the conformal prediction framework, our method guarantees:

(i) A shape-adaptive predictive distribution that captures complex distributional features.

(ii) Finite-sample marginal coverage.

$$\mathbb{P}\{Y_* \in \Gamma_{1-\alpha}(X_*)\} \geq 1 - \alpha.$$

Core Idea: Local PIT Transformation

Idea: Instead of one global shape $\hat{G}$, build a *local* shape $\hat{F}_x$ for each test point x.

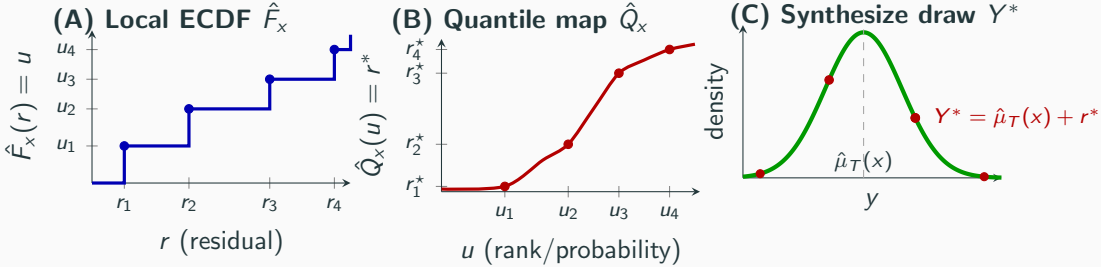
1. Compute residuals: $R_j = Y_j - \hat{\mu}_{\mathcal{T}}(X_j)$ for $j \in \mathcal{T}$.
2. Weight nearby residuals using **kernel weights** $w_j(x) = K_h(\,\|X_j - x\|\,)$
3. Construct a **local empirical CDF** ($\hat{F}_x$) and its **quantile map** ($\hat{Q}_x$).

$$\hat{F}_x(t) = \frac{\sum_{j\in\mathcal{T}} w_j(x)\,\mathbb{I}\{R_j \le t\}}{\sum_{j\in\mathcal{T}} w_j(x)}, \qquad \hat{Q}_x(u) = \inf\{\, t : \hat{F}_x(t) \ge u \,\}.$$

4. Draw predictive samples (Y^*) from this quantile map.

This captures complex local characteristics, producing more precise predictions at the test point.

PAL-PIT: Illustration of the Mechanism



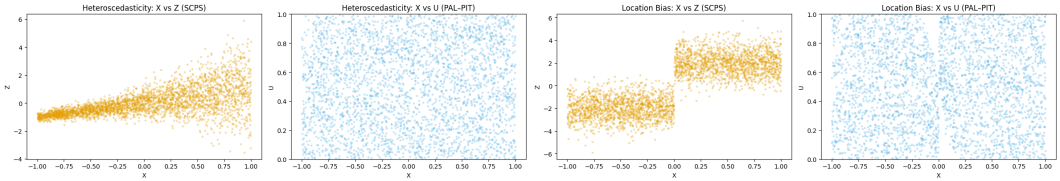
Build local shape from training residuals.

Invert the local ECDF to get a rank-to-residual map ($\hat{Q}_x$ is the generalized inverse of $\hat{F}_x$).

Sample a calibrated rank U, map to $r^* = \hat{Q}_x(U)$, add back to center $\hat{\mu}_T(x)$.

Toy Simulation: Removing X-Dependence

We compare the SCPS score $Z = (Y - \hat{\mu})/\hat{\sigma}$ with the PAL-PIT score $U = \hat{F}_x(Y - \hat{\mu})$.
The plots show Z retains strong dependence on X (left plots), while U successfully removes it (right plots), making it a valid basis for a local distribution.



Heteroscedasticity: X–Z (SCPS) vs. X–U (PAL–PIT) Location Bias: X–Z (SCPS) vs. X–U (PAL–PIT)

Scenario	dCor(X,Z)	dCor(X,U)
Heteroscedasticity	0.6251	**0.0421**
Location bias	0.8206	**0.0368**
Shape mismatch	0.6589	**0.0353**

PAL–PIT shows strictly smaller dependence across all scenarios.

Why Remove the X-Dependence?

1. Directly estimating $F_{Y|X}(\cdot \mid x)$ is data–sparse near x (poor identification).
2. Create a transformed target $Z = T_x(Y)$ that removes the influence of X.
3. If $Z \perp X$, then $F_{Z|X}(t \mid x) = F_Z(t)$ and

$$F_{Y|X}(y \mid x) = F_Z\big(T_x(y)\big).$$

   This allows us to use all samples to estimate F_Z, and simply recover $F_{Y|X}$ by inversion.
4. If Z still depends on X, pooling is invalid; one must use $F_{Z|X}(\cdot \mid x)$, which defeats the purpose of the transform.

The High-D Challenge

The High-D Challenge for Asymptotic Consistency

For the predictive distribution to converge to the true $F_{Y|X}$, the local estimator $\hat{F}_x$ must converge.

- **Challenge:** In high-dimension d, kernel weights based on Euclidean distance, $K_h(\|X_j - x_*\|)$, fail.
- This is the **curse of dimensionality**: all training points X_j become effectively equidistant from the test point x_*.
- "Closeness" becomes meaningless.

High-D Solution: Projections

Auxiliary Tool: Dimension Reduction

- We use dimension reduction (e.g., SIR, SAVE, PLS, Random Proj.) to find a **low-dimensional subspace** $Z = P^\top X \in \mathbb{R}^q$, where $q \ll d$ and P is a projection matrix.
- We then compute kernel weights using distances in this q-dimensional space:

$$w_j(x) = K_h(\|P^\top X_j - P^\top x\|)$$

Asymptotic Consistency

Theorem (Asymptotic Consistency) [Unofficial]

(A1) **Low Dimensionality:** the residual CDF $F_{R|X}$ depends on x only via a q-dim. projection $z = P^T x$.

$$F_{R|X}(t \mid x) = F_{R|Z}(t \mid z) \quad \forall t, x$$

(A2) **Smoothness:** the projected CDF $F_{R|Z}(t \mid z)$ is s-times continuously differentiable ($s \geq 2$) in $z \in \mathbb{R}^q$.

Under A1 and A2, the convergence rate depends only on the low dimension q and smoothness s:

$$\sup_{t \in \mathbb{R}, x \in \mathcal{X}} |\hat{F}_x(t) - F_x(t)| = O_p\left(\left(\frac{\log n}{n}\right)^{\frac{s}{2s+q}}\right)$$

Implication

This is a *weaker assumption* than standard SDR models. We only need the residual shape to depend on a low-d projection, not the entire conditional distribution $Y \mid X$.

PAL-PIT: Mechanism

1. **Split & Fit Center:** Split data into Training $\mathcal{T}$ and Calibration $\mathcal{C}$. Fit base model $\hat{\mu}_{\mathcal{T}}$ on $\mathcal{T}$ only.
2. **Get Residuals:**
   - Compute training residuals $\{R_j = Y_j - \hat{\mu}_{\mathcal{T}}(X_j)\}_{j \in \mathcal{T}}$ (for shape).
   - Compute calibration residuals $\{r_i^{\mathcal{C}} = Y_i - \hat{\mu}_{\mathcal{T}}(X_i)\}_{i \in \mathcal{C}}$ (for ranks).
3. **Build Local Scorer $\hat{F}_x$:** For a test point x, learn projections P using $\{X_j\}_{j \in \mathcal{T}}$. Define kernel weights in the projected space:

$$w_j(x) = K_h(\|P^\top X_j - P^\top x\|)$$

Build the local CDF $\hat{F}_x(t) = \frac{\sum_{j \in \mathcal{T}} w_j(x) \mathbb{I}\{R_j \leq t\}}{\sum_{j \in \mathcal{T}} w_j(x)}$.

PAL-PIT: Mechanism

4. **Get Calibration Ranks:** Apply the *same* scorer $\hat{F}_x$ to the calibration residuals:

$$U_{cal}(x) = \{\hat{F}_x(r_i^{\mathcal{C}}) \mid i \in \mathcal{C}\}$$

5. **Sample (Predictive Dist.)**
   - Draw rank $U \sim \text{Unif}(U_{cal}(x))$.
   - Invert: $r^* = \hat{Q}_x(U) = \inf\{t : \hat{F}_x(t) \geq U\}$.
   - Synthesize: $Y^* = \hat{\mu}_{\mathcal{T}}(x) + r^*$.

Finite-Sample Guarantee

Theorem (Finite-Sample Marginal Coverage) [Unofficial]

Suppose that the pooled data $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n$ and the test point (X_*, Y_*) are exchangeable.

Then, the PAL-PIT prediction set $\Gamma_{1-\alpha}(x)$ satisfies:

$$\mathbb{P}(Y_* \in \Gamma_{1-\alpha}(X_*)) \geq 1 - \alpha$$

This holds for *any* data distribution and *any* choice of training-only projections P or kernel K_h.

Simulation Setup

Scenarios

- **Heavy tail (Cauchy):** Mismatch in tail behavior.
- **Hetero sigmoid:** Variance of noise changes with x.
- **Local bimodal gate:** Residual shape changes from unimodal to bimodal.
- **Skew flip:** Residual skewness changes sign.

Competitors & Projections

- **SCPS:** Standard split-conformal.
- **CART:** Non-conformal resampling from leaf nodes.
- **PAL-PIT:** Our proposed method.

For $d > 1$, PAL-PIT uses projections (SIR (Li, 1991), SAVE (Cook & Weisberg, 1991), PLS (Abdi, 2010), Random Proj. (Kaski, 1998)) to define locality.

Simulated Data: CRPS / Coverage (d = 1, 200)

Dim	Scenario	PAL–PIT	SCPS	CART
1	Heavy tail (Cauchy)	0.6238 / 0.9025	**0.6018** / 0.9050	0.7188 / –
	Hetero sigmoid	**0.4018** / 0.9075	0.4020 / 0.9050	0.4649 / –
	Local bimodal gate	**0.3870** / 0.9000	0.3977 / 0.8950	0.4653 / –
	Skew flip	**0.2418** / 0.9075	0.2440 / 0.9050	0.2898 / –
200	Heavy tail (Cauchy)	**0.7981** / 0.9475	1.0496 / 0.9000	1.4191 / –
	Hetero sigmoid	**0.4559** / 0.8800	0.4723 / 0.9025	1.0108 / –
	Local bimodal gate	**0.3980** / 0.9050	0.4224 / 0.9275	0.9831 / –
	Skew flip	**0.2714** / 0.9125	0.2803 / 0.9350	0.8059 / –

Simulated Data: CRPS / Coverage (d = 500, 1000)

Dim	Scenario	PAL–PIT	SCPS	CART
500	Heavy tail (Cauchy)	**1.0837** / 0.9300	8.0667 / 0.8800	1.6741 / –
	Hetero sigmoid	**0.4511** / 0.9025	0.5148 / 0.9050	0.9926 / –
	Local bimodal gate	**0.4455** / 0.8950	0.5336 / 0.9000	1.0362 / –
	Skew flip	**0.2824** / 0.8725	0.3274 / 0.8850	0.8814 / –
1000	Heavy tail (Cauchy)	1.8395 / 0.9500	9.9842 / 0.9200	**1.4240** / –
	Hetero sigmoid	**0.4652** / 0.9250	0.7045 / 0.8900	1.0245 / –
	Local bimodal gate	**0.4766** / 0.9125	0.7076 / 0.9150	0.9852 / –
	Skew flip	**0.2923** / 0.8975	0.4115 / 0.8900	0.8819 / –

Real-World Data Setup

Datasets

- **CT Slice:** Body scan slice location ($d = 385$)
- **BlogFeedback:** Blog post engagement ($d = 280$)
- **YearPred.MSD:** Song release year ($d = 90$)
- **Superconduct.:** Material critical temperature ($d = 81$)

Protocol & Competitors

- **Protocol:** 80/20 train/test split, 50/50 train/calibration split within training.
- **Competitors:**
  - **SCPS:** Standard global residual pooling.
  - **CART:** Non-conformal leaf resampling
  - **PAL-PIT:** Our method with learned projections.

Real-World Data Analysis: CRPS / Coverage

Dataset	PAL-PIT	SCPS	CART
CT Slice	**1.400** / 0.933	2.892 / 0.923	2.469 / –
BlogFeedback	5.483 / 0.903	6.132 / 0.890	**4.619** / –
YearPred.MSD	**4.919** / 0.900	5.608 / 0.880	5.214 / –
Superconduct.	**6.217** / 0.940	7.889 / 0.895	6.353 / –

Local rank transport yields better predictive distributions (lower CRPS) while maintaining marginal coverage.

Summary

- Replace global residual usage with an x-local empirical distribution constructed by kernel-weighted samples after projection.
- Invert the local empirical CDF to get a quantile map; draw realistic predictive samples for the test point.
- Use a training-only scoring map and evaluate both calibration and test on that scoring map to retain marginal coverage.

Key References

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Thank You

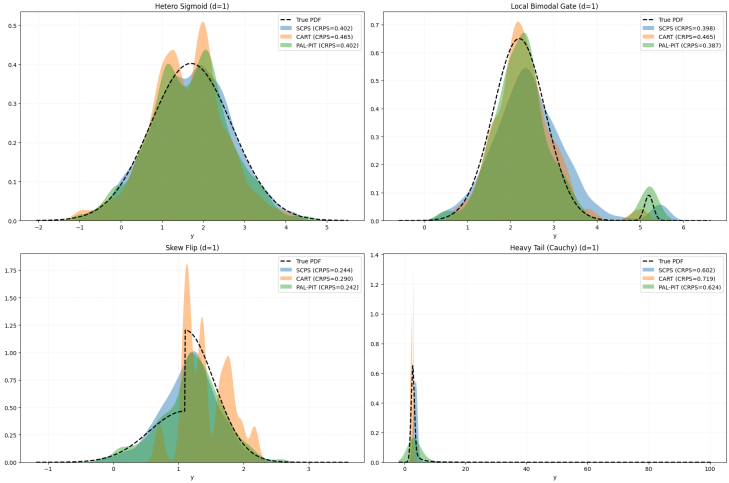
Thank You

Supplementary: PAL-PIT Mechanism (Prediction Set)

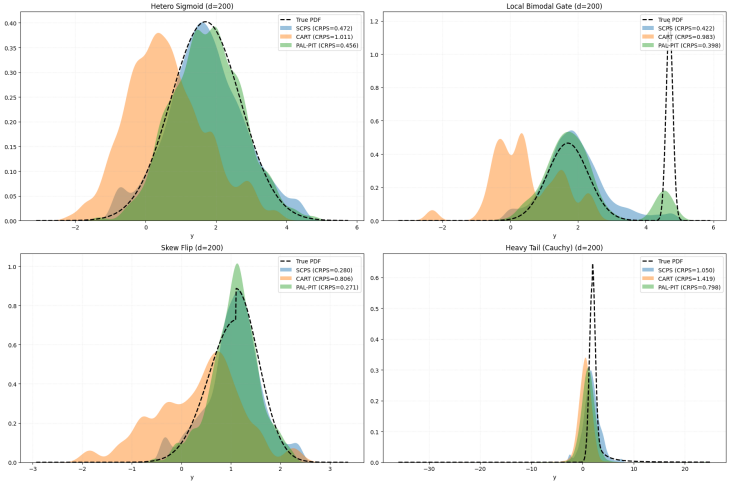
1. **Split & Fit Center:** Split data into Training ($\mathcal{T}$) and Calibration ($\mathcal{C}$). Fit base model $\hat{\mu}_{\mathcal{T}}$ on $\mathcal{T}$ only.
2. **Get Residuals:** $\{R_j = Y_j - \hat{\mu}_{\mathcal{T}}(X_j)\}_{j \in \mathcal{T}}$, $\{r_i^{\mathcal{C}} = Y_i - \hat{\mu}_{\mathcal{T}}(X_i)\}_{i \in \mathcal{C}}$.
3. **Build Local Scorer $\hat{F}_x$:** For a test point x, learn projections P using $\{X_j\}_{j \in \mathcal{T}}$. Build the local CDF.
4. **Get Calibration Ranks:** Apply the *same* score $\hat{F}_x$ to the calibration residuals
5. **Sample (Predictive Dist.)**
   - Draw rank $U \sim \text{Unif}(U_{cal}(x))$.
   - Invert: $r^* = \hat{Q}_x(U) = \inf\{t : \hat{F}_x(t) \geq U\}$.
   - Synthesize: $Y^* = \hat{\mu}_{\mathcal{T}}(x) + r^*$.
6. **Form Prediction Interval ($1 - \alpha$ Coverage):**
   - Let $N = |\mathcal{C}|$. Find ranks τ_{lo} and τ_{hi} as the $\lfloor (N+1)\alpha/2 \rfloor$-th and $\lceil (N+1)(1-\alpha/2) \rceil$-th ordered values from $U_{cal}(x)$.
   - Invert these ranks using the *same* local quantile map $\hat{Q}_x$:

$$\Gamma_{1-\alpha}(x) = \left[\hat{\mu}_{\mathcal{T}}(x) + \hat{Q}_x(\tau_{lo}),\ \hat{\mu}_{\mathcal{T}}(x) + \hat{Q}_x(\tau_{hi})\right].$$

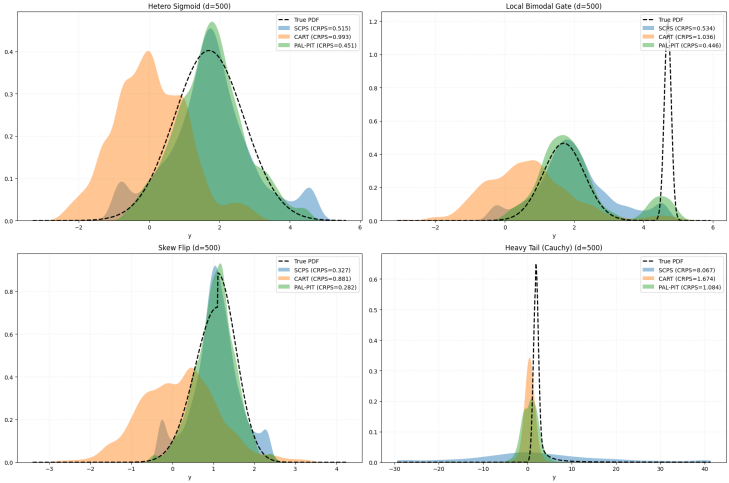
Supplementary: Density Slices (d=1)



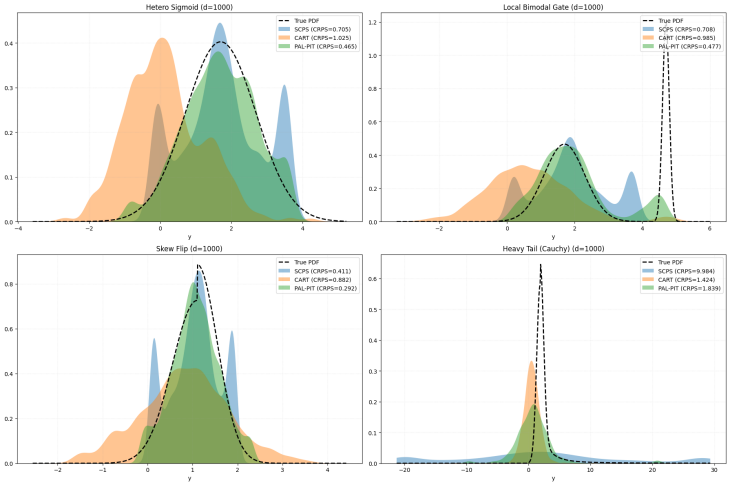
Supplementary: Density Slices (d=200)



Supplementary: Density Slices (d=500)



Supplementary: Density Slices (d=1000)



Supplementary: Dimension Reduction Methods

PAL-PIT uses training-only projections to define locality in high dimensions ($d \gg 1$). These methods find a low-dimensional subspace $Z = P^\top X \in \mathbb{R}^q$.

- **SIR (Sliced Inverse Regression), Li (1991):**
  - *How:* Finds directions based on the **mean** of X in slices of Y ($E[X|Y]$).
  - *When:* Good for finding linear trends.
- **SAVE (Sliced Average Variance Estimation), Cook & Weisberg (1991):**
  - *How:* Finds directions based on the **variance** of X in slices of Y ($Var(X|Y)$).
  - *When:* More robust; can find symmetric relationships (e.g., $Y = X_1^2$) that SIR misses.
- **PLS (Partial Least Squares), Abdi (2010):**
  - *How:* Finds directions that maximize the **covariance** between X and Y.
- **Random Projection, Kaski (1998):**
  - *How:* Uses a random matrix P. Data-agnostic.
  - *When:* Computationally very cheap; surprisingly effective in high-d.